

# Customer Sales Behaviors Analysis

## Exploratory Data Analysis Using Python

### Project Overview

This project analyze behavior of customer from past three thousand nine hundred transactions records across various product categories. The goal is to uncover insights into spending patterns, customers segments, and subscription status to guide strategic business solutions

### Dataset Summary:

Rows = 3900

Columns = 18

Key features:

Customers Demographics:

Age, Gender, Location, Subscription Status

Purchase Details:

Item Purchased, Category, Purchase Amount, Season, Size, Color

Shopping behavior:

Discount Applied, Promo Code, Previous Purchases, Frequency of Purchase, Review Rating, Shipping Type

Missing Data:

37 values in Rating Columns

### Exploratory Data Analysis Using Python

We began with data preparation and cleaning in python:

**Data loading:** imported the dataset using pandas

**Initial Exploration:** Used `df.head()`, `df.info()` to check structure and `df.describe()` for summary statistics

`df.head()`

```
Data loaded successfully!
   Customer ID  Age  Gender  ...  Previous Purchases  Payment Method  Frequency of Purchases
0            1   55   Male  ...             14          Venmo          Fortnightly
1            2   19   Male  ...              2           Cash          Fortnightly
2            3   50   Male  ...             23  Credit Card           Weekly
3            4   21   Male  ...             49         PayPal           Weekly
4            5   45   Male  ...             31         PayPal           Annually

[5 rows x 18 columns]
```

`df.info()`

```
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null  int64
1   Age                                  3900 non-null  int64
2   Gender                              3900 non-null  str
3   Item Purchased                       3900 non-null  str
4   Category                             3900 non-null  str
5   Purchase Amount (USD)                3900 non-null  int64
6   Location                             3900 non-null  str
7   Size                                 3900 non-null  str
8   Color                                3900 non-null  str
9   Season                               3900 non-null  str
10  Review Rating                        3863 non-null  float64
11  Subscription Status                  3900 non-null  str
12  Shipping Type                       3900 non-null  str
13  Discount Applied                    3900 non-null  str
14  Promo Code Used                     3900 non-null  str
15  Previous Purchases                   3900 non-null  int64
16  Payment Method                      3900 non-null  str
17  Frequency of Purchases               3900 non-null  str
dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB
```

`df.describe()`

|        | Customer ID | Age         | Gender | ... | Previous Purchases | Payment Method | Frequency of Purchases |
|--------|-------------|-------------|--------|-----|--------------------|----------------|------------------------|
| count  | 3900.000000 | 3900.000000 | 3900   | ... | 3900.000000        | 3900           | 3900                   |
| unique | NaN         | NaN         | 2      | ... | NaN                | 6              | 7                      |
| top    | NaN         | NaN         | Male   | ... | NaN                | PayPal         | Every 3 Months         |
| freq   | NaN         | NaN         | 2652   | ... | NaN                | 677            | 584                    |
| mean   | 1950.500000 | 44.068462   | NaN    | ... | 25.351538          | NaN            | NaN                    |
| std    | 1125.977353 | 15.207589   | NaN    | ... | 14.447125          | NaN            | NaN                    |
| min    | 1.000000    | 18.000000   | NaN    | ... | 1.000000           | NaN            | NaN                    |
| 25%    | 975.750000  | 31.000000   | NaN    | ... | 13.000000          | NaN            | NaN                    |
| 50%    | 1950.500000 | 44.000000   | NaN    | ... | 25.000000          | NaN            | NaN                    |
| 75%    | 2925.250000 | 57.000000   | NaN    | ... | 38.000000          | NaN            | NaN                    |
| max    | 3900.000000 | 70.000000   | NaN    | ... | 50.000000          | NaN            | NaN                    |

To check number of missing values we use `print(df.isnull().sum())` by that we know that how many missing values are there.

```

[11 rows x 18 columns]
Missing values in each column:
Customer ID          0
Age                  0
Gender               0
Item Purchased       0
Category             0
Purchase Amount (USD) 0
Location            0
Size                0
Color               0
Season              0
Review Rating       37
Subscription Status  0
Shipping Type       0
Discount Applied    0
Promo Code Used     0
Previous Purchases  0
Payment Method      0
Frequency of Purchases 0
dtype: int64
PS D:\Desktop\Visual Studio Code_Python>

```

To populate these null values we use that code to populate it according to its category for clothing we use median of cloths and same as it is for other category

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x:
x.fillna(x.median()))
```

by that code our review rating missing values populate by the median of their respective category

```
print("Data after filling missing values:")
print(df.isnull().sum())
```

That code is used to verify missing values either they are populated or not

This script is use to perform many tasks

- 1- Convert uppercase column name to lower case
- 2- Replace (space) with underscore ( \_ )
- 3- To rename name of different columns
- 4- Last print command is used to print all the column names

```
df.columns = df.columns.str.lower()

df.columns = df.columns.str.replace(' ', '_') df =
df.rename(columns={'promo_code_used': 'promo_code',})
df=df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
df=df.rename(columns={'frequency_of_purchases': 'purchase_frequency'})
) print(df.columns)
```

Data after filling missing values

```
Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season          0
Review Rating    0
Subscription Status  0
Shipping Type    0
Discount Applied 0
Promo Code Used  0
Previous Purchases 0
Payment Method   0
Frequency of Purchases 0
dtype: int64
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code', 'previous_purchases',
      'payment_method', 'purchase_frequency'],
      dtype='str')
```

For categorizing age

```
df['age_group'] = pd.cut(
    df['age'], [18, 25, 40, 60, 100],
    labels=['young_adults', 'adults', 'middle-aged', 'seniors']
) print(df[['age',
'age_group']].head(10))
```

This script is used to cut age in to different category first we create new column name age\_group then use that function pd.cut() and then write name of that column where data is available which we want to cut in to different categories and then write label for that and print both columns

|   | age | age_group    |
|---|-----|--------------|
| 0 | 55  | middle-aged  |
| 1 | 19  | young_adults |
| 2 | 50  | middle-aged  |
| 3 | 21  | young_adults |
| 4 | 45  | middle-aged  |
| 5 | 46  | middle-aged  |
| 6 | 63  | seniors      |
| 7 | 27  | adults       |
| 8 | 26  | adults       |
| 9 | 57  | middle-aged  |

The `frequency_mapping` dictionary is created to convert the text values in the `purchase_frequency` column into their equivalent number of days. Each purchase frequency is assigned a numeric value. For example:

- Weekly = 7 days
- Fortnightly = 14 days
- Monthly = 30 days
- Quarterly = 90 days
- Annually = 365 days

The `.map()` function is then applied to the `purchase_frequency` column. It looks up each value in the dictionary and replaces it with the corresponding number of days. The converted values are stored in a new column called `purchase_frequency_days`, while the original `purchase_frequency` column remains unchanged.

Finally, the `head(10)` function is used to display the first 10 rows of both the original and the newly created column to verify that the mapping has been performed correctly.

|   | purchase_frequency_days | purchase_frequency |
|---|-------------------------|--------------------|
| 0 | 14.0                    | Fortnightly        |
| 1 | 14.0                    | Fortnightly        |
| 2 | 7.0                     | Weekly             |
| 3 | 7.0                     | Weekly             |
| 4 | 365.0                   | Annually           |
| 5 | 7.0                     | Weekly             |
| 6 | 90.0                    | Quarterly          |
| 7 | 7.0                     | Weekly             |
| 8 | 365.0                   | Annually           |
| 9 | 90.0                    | Quarterly          |

```
print(df[['discount_applied', 'promo_code']].head(10))
df['discount_applied'] == df['promo_code'].all() df =
df.drop('promo_code', axis=1) print(df.columns)
```

This script is used to remove duplicate column promo code because promo code and discount applied is same so we remove one column to make our analysis easy.

## Data Analysis Using SQL:

Total revenue generated by gender:

|   | gender<br>text | total_cost<br>numeric |
|---|----------------|-----------------------|
| 1 | Female         | 75191                 |
| 2 | Male           | 157890                |

Customers with more purchasing with discount applied:

|   | customer_id<br>bigint | purchase_amount<br>bigint | discount_applied<br>text |
|---|-----------------------|---------------------------|--------------------------|
| 1 | 2                     | 64                        | Yes                      |
| 2 | 3                     | 73                        | Yes                      |
| 3 | 4                     | 90                        | Yes                      |
| 4 | 7                     | 85                        | Yes                      |
| 5 | 9                     | 97                        | Yes                      |
| 6 | 12                    | 68                        | Yes                      |
| 7 | 13                    | 72                        | Yes                      |
| 8 | 16                    | 81                        | Yes                      |
| 9 | 20                    | 90                        | Yes                      |

Top 5 products with highest average review rating:

|   | item_purchased<br>text | average<br>numeric |
|---|------------------------|--------------------|
| 1 | Gloves                 | 3.86               |
| 2 | Sandals                | 3.84               |
| 3 | Boots                  | 3.82               |
| 4 | Hat                    | 3.80               |
| 5 | Skirt                  | 3.78               |

Average revenue by standard or express shipping:

|   | shipping_type<br>text | average<br>numeric |
|---|-----------------------|--------------------|
| 1 | Standard              | 58.46              |
| 2 | Express               | 60.48              |

Revenue by Subscribers or non:

|   | subscription_status<br>text | total_customers<br>bigint | total_revenue<br>numeric | average_revenue<br>numeric |
|---|-----------------------------|---------------------------|--------------------------|----------------------------|
| 1 | No                          | 2847                      | 170436                   | 59.87                      |
| 2 | Yes                         | 1053                      | 62645                    | 59.49                      |

Top 5 Item purchases by discount rate:

|   | item_purchased<br>text | discount_rate<br>numeric |
|---|------------------------|--------------------------|
| 1 | Hat                    | 50.00                    |
| 2 | Sneakers               | 49.00                    |
| 3 | Coat                   | 49.00                    |
| 4 | Sweater                | 48.00                    |
| 5 | Pants                  | 47.00                    |

Customer segmentation into loyal, new and returning :

|   | customer_segment<br>text | number_of_customers<br>bigint |
|---|--------------------------|-------------------------------|
| 1 | loyal                    | 3116                          |
| 2 | new                      | 83                            |
| 3 | Returning                | 701                           |



### Top 3 product by Category :

|    | item_rank<br>bigint | category<br>text | item_purchased<br>text | total_customers<br>bigint |
|----|---------------------|------------------|------------------------|---------------------------|
| 1  | 1                   | Accessori...     | Jewelry                | 171                       |
| 2  | 2                   | Accessori...     | Sunglasses             | 161                       |
| 3  | 3                   | Accessori...     | Belt                   | 161                       |
| 4  | 1                   | Clothing         | Blouse                 | 171                       |
| 5  | 2                   | Clothing         | Pants                  | 171                       |
| 6  | 3                   | Clothing         | Shirt                  | 169                       |
| 7  | 1                   | Footwear         | Sandals                | 160                       |
| 8  | 2                   | Footwear         | Shoes                  | 150                       |
| 9  | 3                   | Footwear         | Sneakers               | 145                       |
| 10 | 1                   | Outerwear        | Jacket                 | 163                       |
| 11 | 2                   | Outerwear        | Coat                   | 161                       |

### Repeat buyers are Subscriber or not :

|   | subscription_status<br>text | total_customers<br>bigint |
|---|-----------------------------|---------------------------|
| 1 | No                          | 2518                      |
| 2 | Yes                         | 958                       |

### Revenue Contribution by age group

|   | age_group<br>text | total_revenue<br>numeric |
|---|-------------------|--------------------------|
| 1 | middle-aged       | 89445                    |
| 2 | adults            | 65842                    |
| 3 | seniors           | 43164                    |
| 4 | young_adul...     | 30491                    |
| 5 | [null]            | 4139                     |

Developing Dashboard using power bi



If you are explaining this dashboard in a **theory paper** or **viva**, you can write it like this:

The **Customers Sales Dashboard** provides a visual summary of customer purchasing behavior, revenue, and sales performance. It helps users quickly analyze important business metrics through interactive charts and KPIs.

## 1. Dashboard Title

### Customers Sales Dashboard

The title indicates that the dashboard is designed to analyze customer sales data and business performance.

## 2. Filters (Slicers)

The filters on the left side allow users to interact with the dashboard and view data according to specific conditions.

- ❑ **Gender:** Male, Female
- ❑ **Discount Applied:** Yes, No ❑ **Subscription Status:** Yes, No
- **Payment Method:** Bank Transfer, Cash, Credit Card, Debit Card, PayPal, Venmo
- **Category:** Clothing, Accessories, Footwear, Outerwear

These filters update all charts and KPIs dynamically based on the selected criteria.

### 3. KPI Cards

#### (a) Number of Customers

**Value:** 3.9K

This KPI shows the total number of customers included in the dataset.

#### (b) Average Purchase

**Value:** \$59.76

This KPI represents the average amount spent by customers on each purchase. c)

#### Average Review

**Value:** 3.75

This KPI displays the average customer rating or review score for purchased products.

### 4. Percentage of Customers by Subscription Status (Donut Chart)

This donut chart shows the distribution of customers based on whether they have a subscription.

- **Subscribed (Yes):** 1.05K customers
- **Not Subscribed (No):** 2.85K customers

It helps identify how many customers are subscribed compared to non-subscribers.

### 5. Revenue by Gender (Column Chart)

This chart compares total revenue generated by male and female customers.

- **Male:** Approximately **158K**
- **Female:** Approximately **75K**

The chart indicates that male customers generated more revenue than female customers.

### 6. Revenue by Category (Bar Chart)

This chart displays revenue generated by each product category.

- Clothing generates the highest revenue.

- Accessories is the second-highest category.
- Footwear generates moderate revenue.
- Outerwear contributes the least revenue.

This helps identify the best-performing product categories.

## 7. Sales by Age Group (Horizontal Bar Chart)

This chart compares the number of sales across different age groups.

- Middle-aged customers made the highest number of purchases.
- Adults made the second-highest number of purchases. □ Seniors made the fewest purchases.

It helps understand which age group contributes the most sales.

## 8. Revenue by Age Group (Horizontal Bar Chart)

This chart shows the total revenue generated by each age group.

- Middle-aged customers contribute the highest revenue. □ Adults generate the second-highest revenue.

This helps businesses identify the most valuable customer age group.

## Summary

The dashboard provides a comprehensive overview of customer sales performance by combining KPIs, charts, and interactive filters. It enables users to analyze customer behavior, compare revenue across gender, categories, and age groups, monitor subscription status, and support datadriven business decisions. Interactive slicers make the dashboard dynamic by allowing users to filter the analysis based on gender, payment method, subscription status, discount application, and product category.